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PY360: Ethics in Technology

Assignment 04 — From Alphabets to Algorithms: Ethics in the Information Age

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Part 1: From Alpha to Hypertext: Early Milestones in Information Storage and Retrieval

The story of information storage and retrieval is, in many ways, the story of human civilization itself. Long before the digital era, human beings sought ways to preserve knowledge beyond the limits of individual memory. What follows is an overview of the key milestones that led from the earliest writing systems to the revolutionary concept of hypertext.

For thousands of years, tribal peoples relied on oral tradition to preserve their stories, myths, and cultural knowledge. Elders passed down information through spoken word and ritual reenactment. While effective within small communities, oral tradition was inherently limited — knowledge could be lost if a storyteller died before passing it on (Quinn, 2020).

The first major leap in information storage came roughly six thousand years ago with the invention of writing, primarily in the river valleys of Mesopotamia, Egypt, and the Yellow River basin in China. Early writing systems were ideographic, meaning they used symbols to represent whole words or concepts. Egyptian hieroglyphics are perhaps the most well-known example. However, ideographic systems required users to memorize hundreds or even thousands of symbols, which limited literacy to a small elite class (Quinn, 2020).

The development of the Greek alphabet represented a transformative advancement. By reducing language to a small set of phonetic symbols — alpha, beta, gamma, delta, and so on — the Greeks made reading and writing far more accessible. These symbols eventually evolved into the familiar letters of the modern English alphabet. The alphabet was revolutionary because it dramatically lowered the barrier to literacy, enabling broader segments of society to record, store, and retrieve information (Quinn, 2020).

With the rise of written language came the need for durable storage media. Early civilizations used papyrus scrolls, which served well for a time but were vulnerable to dry rot and fire. In the fourth century, the codex emerged as a superior alternative. A codex consisted of bound rectangular pages made from sheepskin or calfskin, with pages arranged in numerical order. This format was more durable than scrolls and, critically, made it far easier to locate a particular passage — an early improvement in information retrieval (Quinn, 2020).

The next major development came when Western civilizations adopted paper, a Chinese invention, which was far cheaper and easier to produce than animal-skin parchment. Paper set the stage for what many consider the single most important innovation in the history of information storage and retrieval: Johannes Gutenberg's printing press, developed around 1436. By combining movable type with a mechanical press, Gutenberg made it possible to produce books rapidly and inexpensively. The printing press democratized access to information in ways that were previously unimaginable. Books proliferated, literacy rates climbed, and new ideas

could spread rapidly across entire continents. The consequences were profound — the printing press helped fuel the Renaissance, the Reformation, and eventually the Enlightenment (Quinn, 2020).

For centuries after Gutenberg, the basic model of information storage and retrieval remained largely the same: knowledge was recorded in printed form and organized in libraries using cataloging systems. However, in the twentieth century, a Harvard computer scientist named Ted Nelson introduced a concept that would fundamentally change how humans interact with stored information. Nelson called his invention hypertext, which he defined as a network of nodes containing information, where links allow readers to visit the nodes in a nonlinear fashion (Quinn, 2020).

Hypertext was revolutionary because it mimicked the associative nature of human memory. Rather than reading information sequentially, from beginning to end as one does with a book, a hypertext user could follow links from one piece of information to another based on relevance and interest. This concept laid the intellectual groundwork for what would eventually become the World Wide Web, web browsing, and search engines — the very tools that define the modern information age (Quinn, 2020).

In summary, the journey from the Greek alphabet to hypertext reflects humanity's enduring drive to store, organize, and access knowledge more effectively. Each milestone — the alphabet, the codex, paper, the printing press, and hypertext — built upon what came before, expanding access to information and reshaping society in the process.

Part 2: Justice Behind the Veil: John Rawls' Theory of Justice

How can a society create rules that are truly fair to everyone? This is the central question that American philosopher John Rawls sought to answer in his influential theory of justice. Rather than relying on tradition, religious authority, or the preferences of those in power, Rawls proposed a thought experiment designed to strip away personal bias and arrive at principles of justice that rational people would universally accept.

Rawls' theory belongs to the tradition of social contract theory, which holds that the rules governing a society should be based on an agreement among its members. However, Rawls recognized a fundamental problem with most social agreements: people tend to advocate for rules that benefit themselves. A wealthy person might argue for low taxes, while a poor person might argue for generous social programs. To overcome this self-interested bias, Rawls designed an ingenious thought experiment (Quinn, 2020).

Rawls asked us to imagine a group of people coming together to establish the fundamental principles of their society. He called this starting point the original position. In the original position, the participants are rational, self-interested individuals who understand general facts about human society — economics, psychology, and politics — but with one crucial restriction: they are placed behind what Rawls called the veil of ignorance (Quinn, 2020).

The veil of ignorance is the most creative and important element of Rawls' theory. Behind this veil, no one knows their own place in society. They do not know whether they will be rich or poor, healthy or disabled, male or female, or a member of a majority or minority group. They do not know their own talents, intelligence, or personal values. The veil strips away every piece of information that might tempt a person to rig the rules in their own favor (Quinn, 2020).

Rawls argued that rational people in the original position, operating behind the veil of ignorance, would agree on two fundamental principles of justice. The first principle holds that each person should have equal right to the most extensive system of basic liberties compatible with a similar system of liberties for all. In simpler terms, everyone gets the same fundamental freedoms — speech, assembly, conscience, and so on — and no one's freedoms can be sacrificed for someone else's benefit (Quinn, 2020).

The second principle addresses social and economic inequalities, and it contains the key concept known as the difference principle. The difference principle acknowledges that perfect economic equality may not be realistic or even desirable. Instead, it holds that social and economic inequalities are acceptable only if they benefit the least advantaged members of society. For example, allowing doctors to earn high salaries might be acceptable if it incentivizes talented people to enter medicine, ultimately improving healthcare for everyone, including the poorest members of society (Quinn, 2020).

To illustrate with a creative example, imagine a group of strangers designing the rules for a new neighborhood. No one knows yet whether they will be the wealthiest homeowner or the person struggling to make rent. Behind this veil of ignorance, it is unlikely that anyone would vote for rules that allow the wealthy to monopolize all shared resources. Instead, they would likely agree on rules ensuring that everyone has access to basic amenities, and that any special advantages granted to some residents ultimately improve conditions for the least fortunate among them.

Rawls' theory has significant implications for technology ethics. When considering policies about internet access, data privacy, or the distribution of technological benefits, Rawls would urge us to ask: Would we support this policy if we did not know whether we would be among the technology haves or the technology have-nots? This perspective provides a powerful lens for evaluating the fairness of policies in the information age.

Part 3: Caught in the Web: Evidence and Expert Perspectives on Internet Addiction

The question of whether Internet addiction is a genuine phenomenon has become increasingly relevant as digital technology permeates virtually every aspect of daily life. Drawing on personal observation and expert analysis from the text, this essay examines evidence that Internet addiction exists and explores the factors that contribute to it.

From personal experience, the signs of Internet addiction are visible in everyday life. Consider the common scenario of sitting down to check email or social media for "just a minute," only to look up and realize that an hour or more has vanished. Many people experience a sense of

anxiety or restlessness when separated from their smartphones, even for short periods. Students in a college classroom, despite knowing they should be focused on a lecture, find themselves compulsively checking notifications. Friends at a dinner table sit across from one another yet interact more with their screens than with each other. These behaviors suggest a compulsive quality to Internet use that goes beyond simple enjoyment or utility.

Perhaps most telling is the experience of attempting to deliberately reduce screen time. Many people who try to limit their Internet use report difficulty following through, a pattern that mirrors the struggle associated with other recognized forms of addiction. The pull to check one more post, watch one more video, or play one more round of a game can feel genuinely difficult to resist, even when the person recognizes that the behavior is unproductive or harmful to their well-being.

Expert opinion presented in the text lends support to these personal observations. Psychiatrist Jerald Block identified three specific variants of Internet addiction: excessive gaming, sexual preoccupations, and excessive e-mail and text messaging. Block's framework suggests that Internet addiction is not a single behavior, but a family of compulsive behaviors enabled by digital technology (Quinn, 2020).

Researcher Kimberly Young has contributed significant insight into the neurological dimension of the problem. Young maintains that the brains of heavy users of digital devices are rewarded through a "high" produced by increased production of dopamine, a neurotransmitter associated with pleasure and reward. This dopamine mechanism is strikingly similar to the neurological processes observed in substance addiction and gambling addiction, providing a biological basis for the claim that Internet addiction is real and not merely a matter of weak willpower (Quinn, 2020).

Several factors contribute to the development of Internet addiction. First, the design of many digital platforms deliberately encourages compulsive use. Features such as infinite scrolling, push notifications, autoplay videos, and variable reward schedules — where the user never knows when the next interesting or gratifying piece of content will appear — mimic the psychological mechanisms of slot machines. These design choices are not accidental; they are engineered to maximize user engagement, which translates directly into advertising revenue for the platform.

Second, the Internet provides an unmatched variety of stimulation. Unlike a single activity such as reading a book or watching television, the Internet offers an essentially infinite array of content and interaction — gaming, social media, news, shopping, video, and more. This variety means that when interest in one activity wanes, a user can instantly switch to another, sustaining engagement for extended periods.

Third, the social dimension of Internet use creates its own form of dependency. Social media platforms foster a sense of obligation to respond to messages, comment on posts, and maintain

an online presence. The fear of missing out, commonly referred to as FOMO, drives many users to check their accounts far more frequently than they otherwise would.

Finally, from a philosophical perspective, the text notes that thinkers like Kant and Mill would place responsibility for excessive Internet use on the individual, since both traditions emphasize personal autonomy and rational self-governance. However, this raises an important counter-question: if platforms are deliberately designed to exploit neurological vulnerabilities, is it entirely fair to place the blame solely on the user? This tension between personal responsibility and corporate accountability remains one of the most pressing ethical questions of the information age (Quinn, 2020).

In conclusion, both personal experience and expert analysis support the claim that Internet addiction is a genuine phenomenon. The combination of deliberate platform design, neurological reward mechanisms, endless content variety, and social pressure creates an environment in which compulsive Internet use can flourish. As digital technology continues to evolve, understanding and addressing Internet addiction will remain an important ethical and public health concern.

Reference

Quinn, M. J. (2020). *Ethics for the information age* (8th ed.). Pearson.